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Nishi et al.

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(54) **LIQUID COLLECTOR**

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(57) **ABSTRACT**

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An embodiment of the present disclosure provides a liquid collector that collects liquid, the liquid collector including: a flow passage member including a flow passage that connects one end and other end of the flow passage member; a contact member that is disposed at the one end of the flow passage member, that includes an elastic material, and that is configured to come into contact with a target from which liquid is to be collected; and a container that receives the liquid discharged from the flow passage at the other end of the flow passage member.

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(52) **U.S. Cl.**

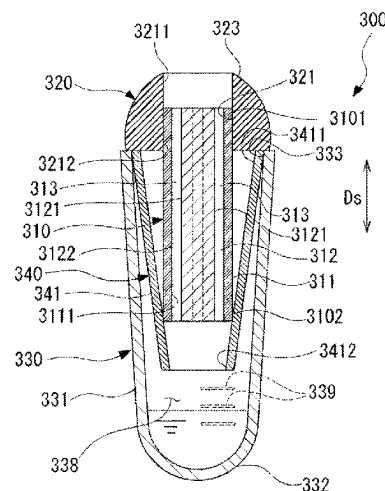
CPC **G01N 1/10** (2013.01); **A61B 2010/0067** (2013.01); **G01N 2001/028** (2013.01)

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18 Claims, 14 Drawing Sheets



(58) **Field of Classification Search**

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See application file for complete search history.

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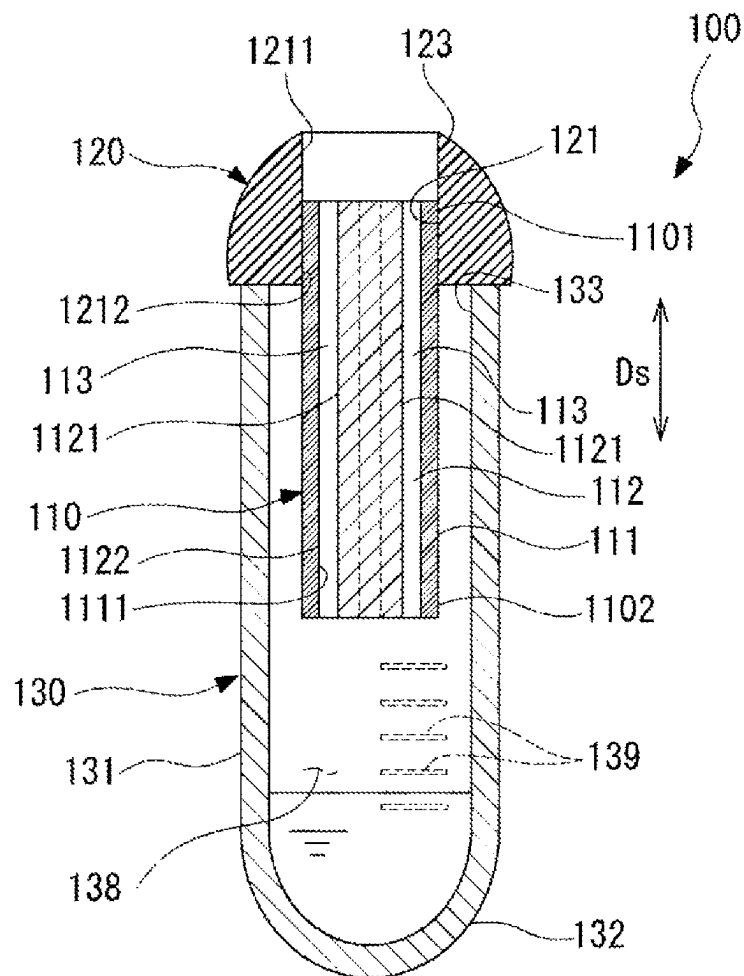


FIG. 1

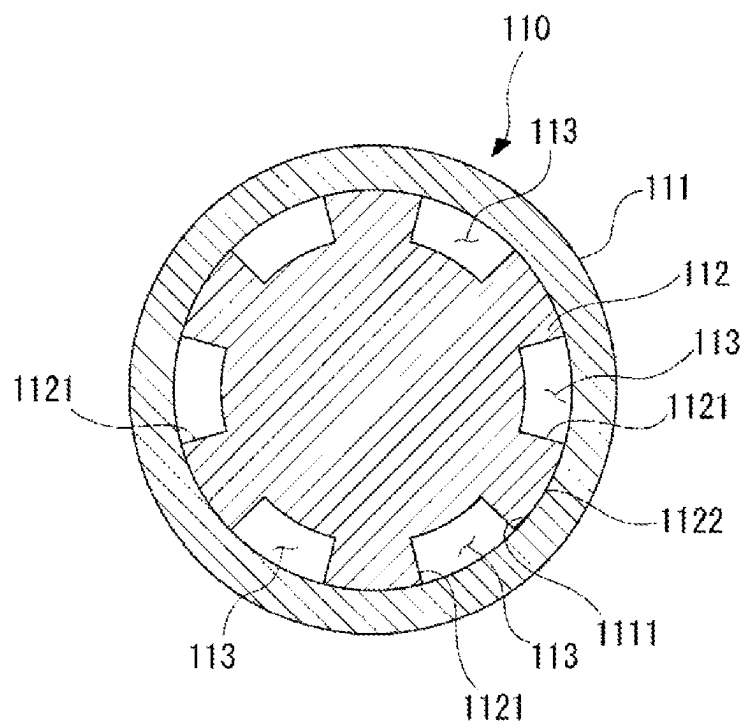


FIG. 2

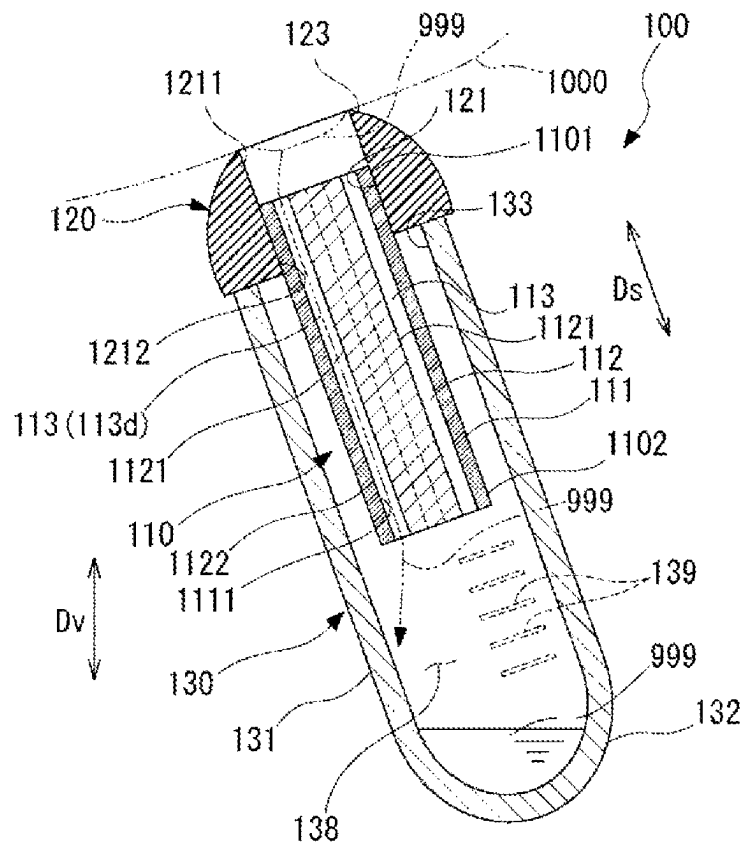


FIG. 3

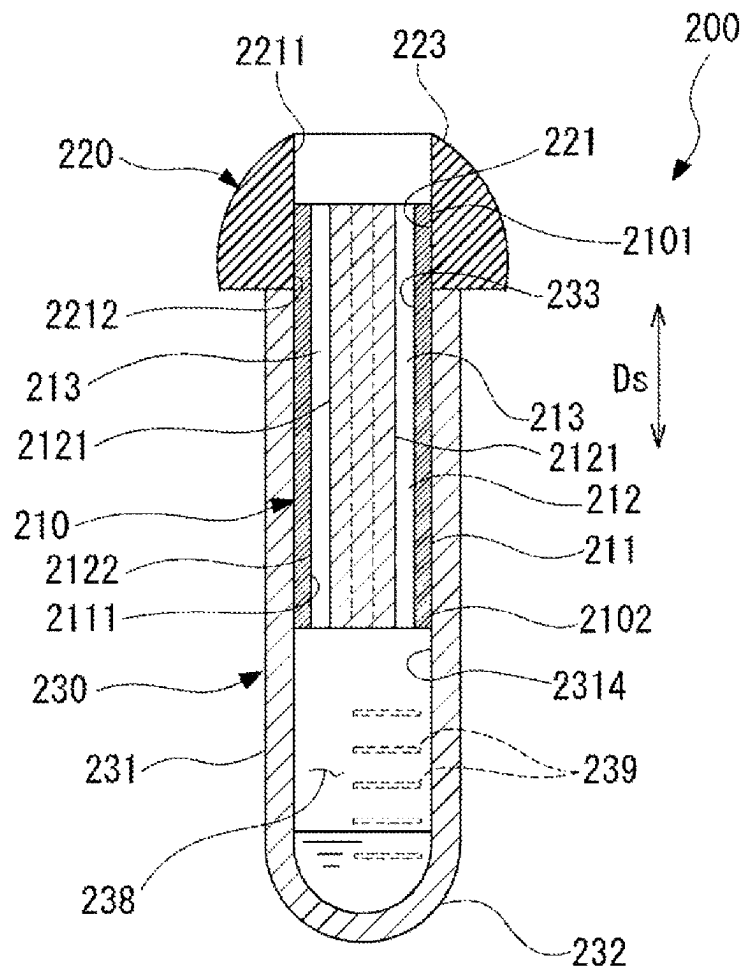


FIG. 4

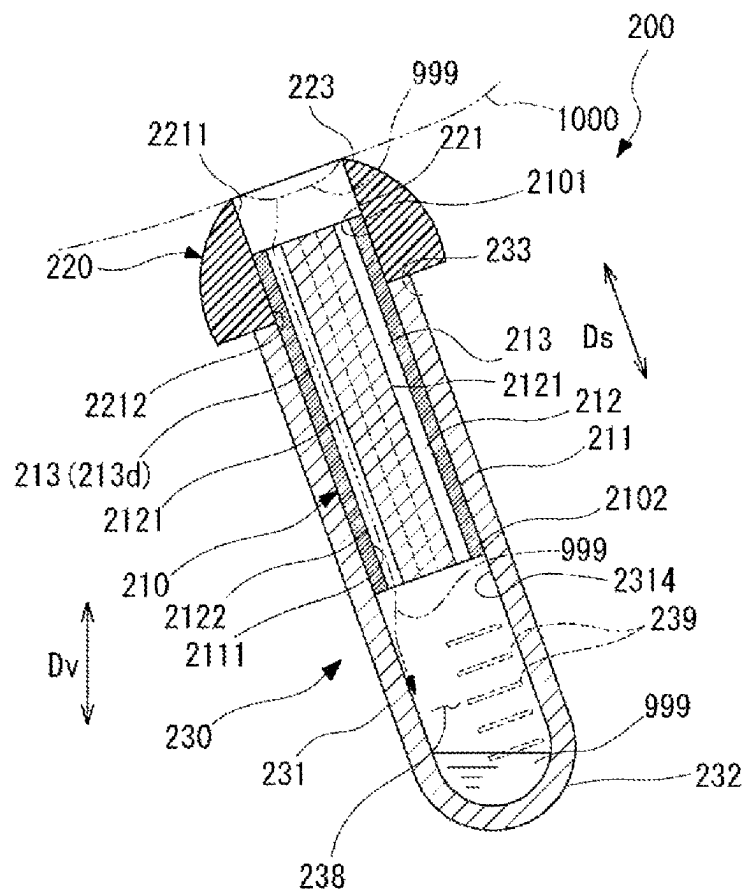


FIG. 5

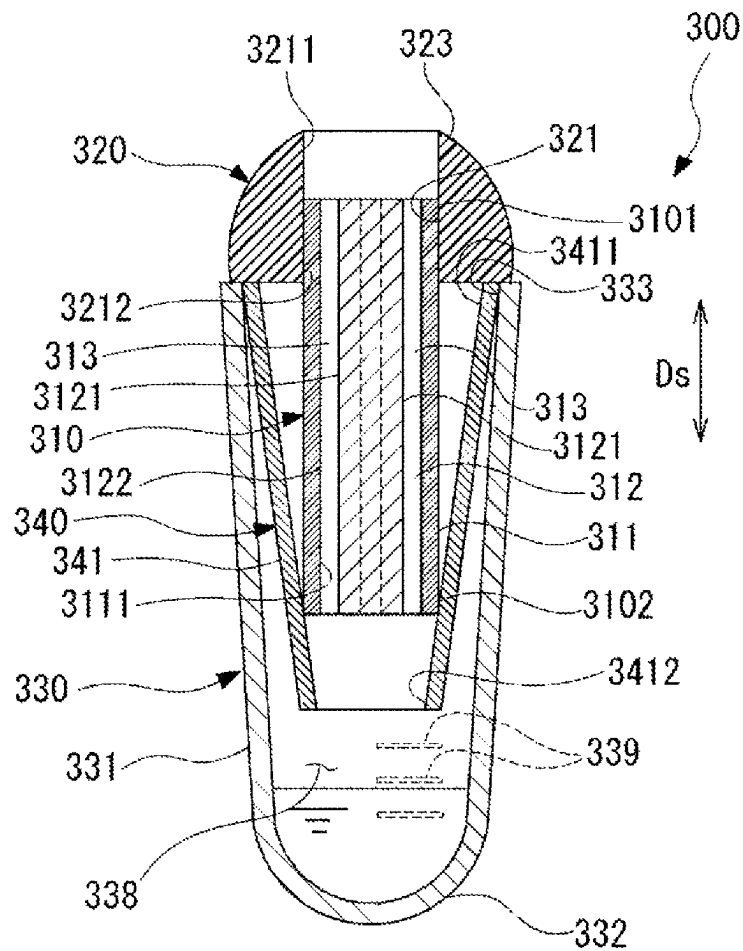


FIG. 6

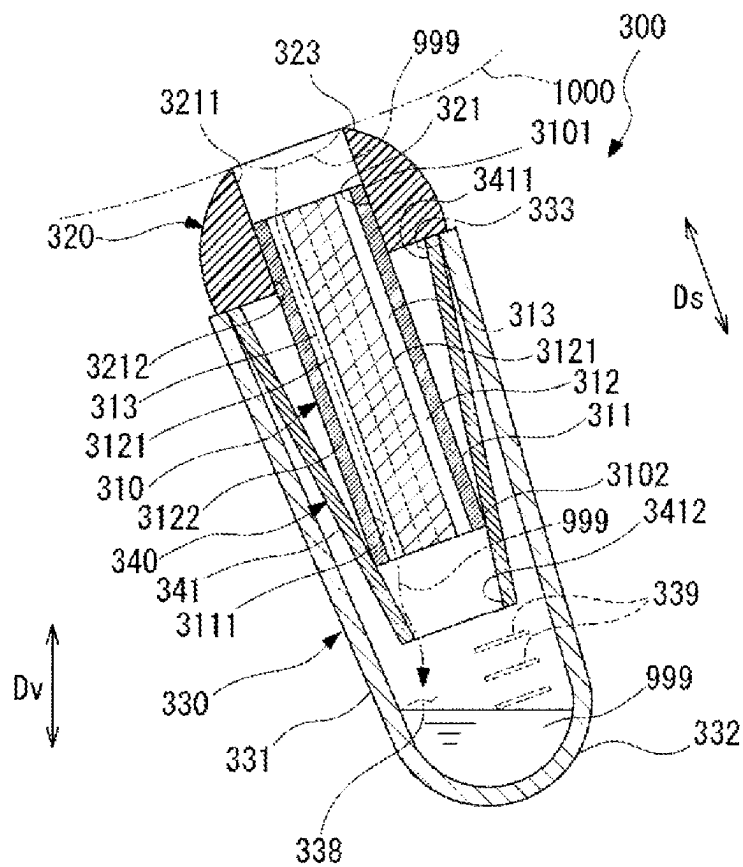


FIG. 7

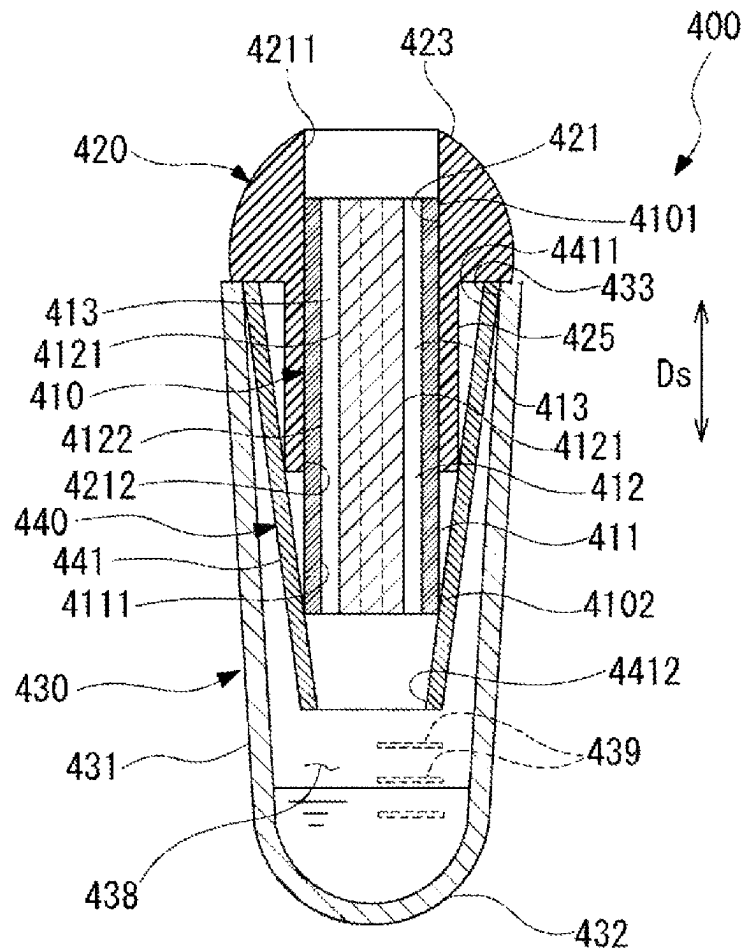


FIG. 8

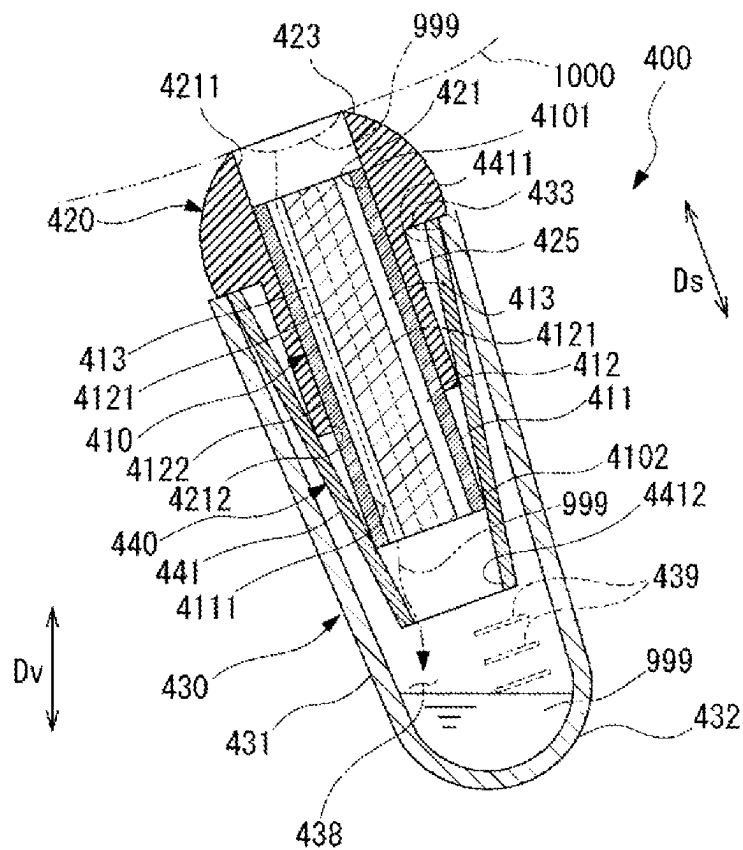


FIG. 9

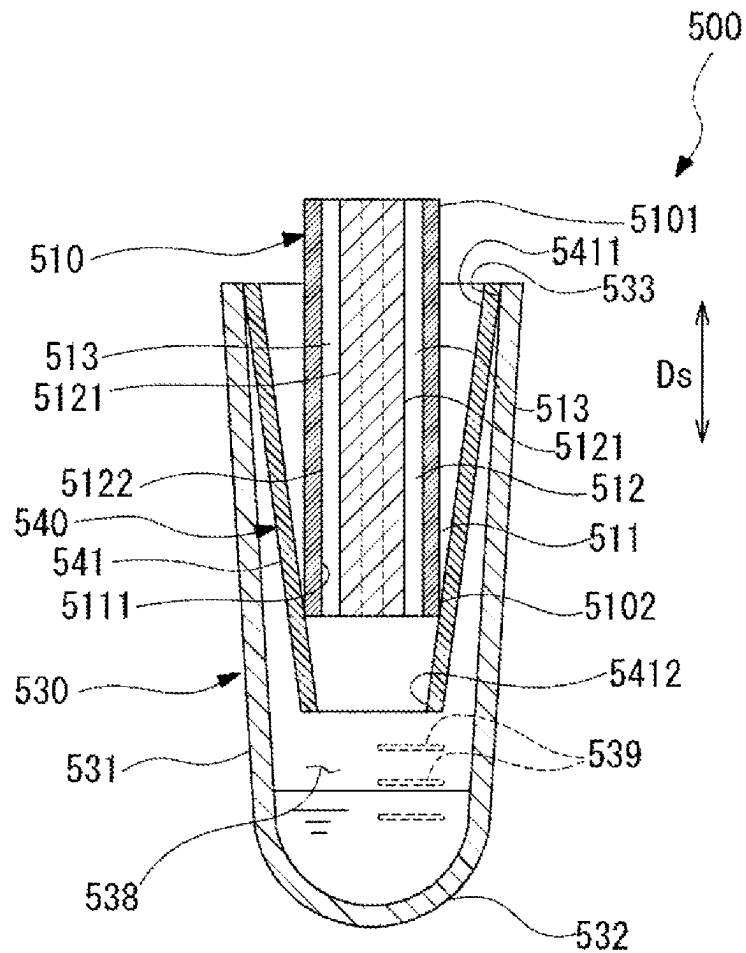


FIG. 10

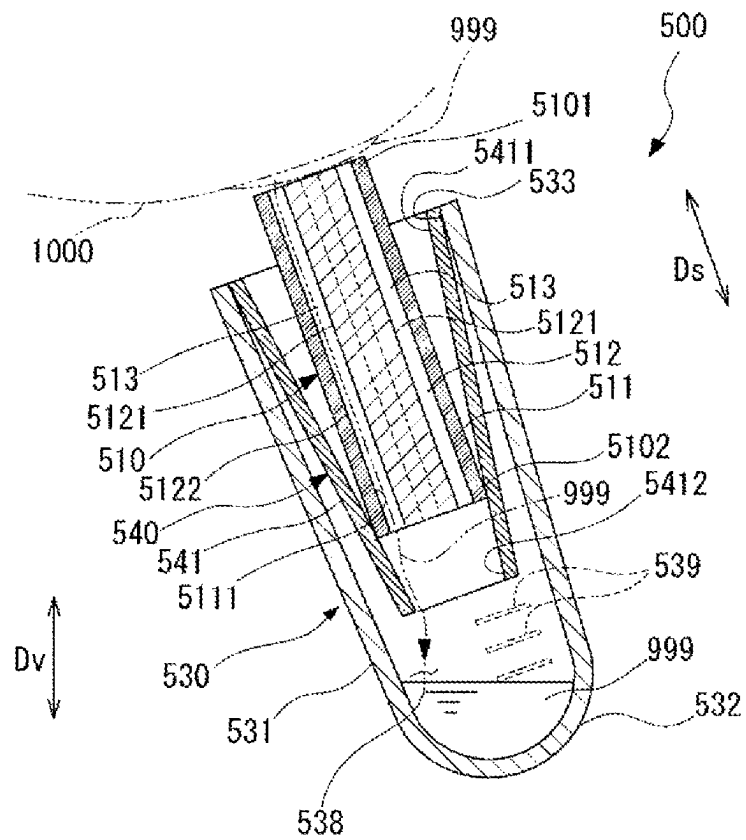


FIG. 11

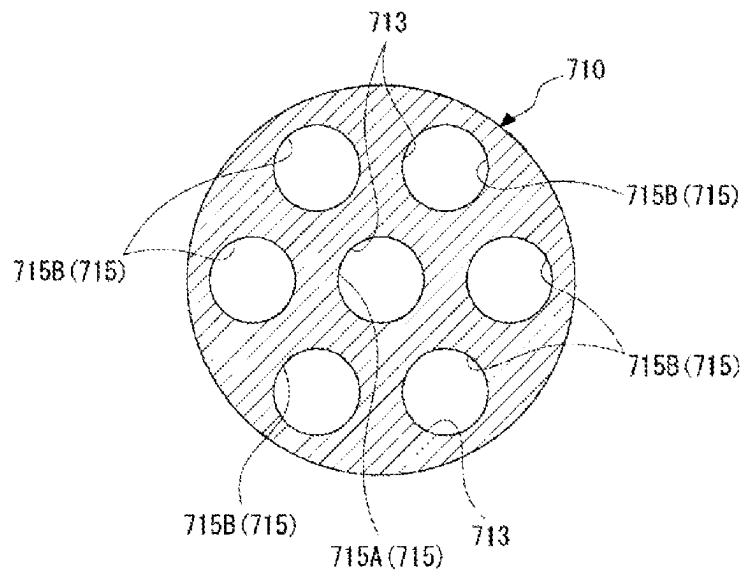


FIG. 12

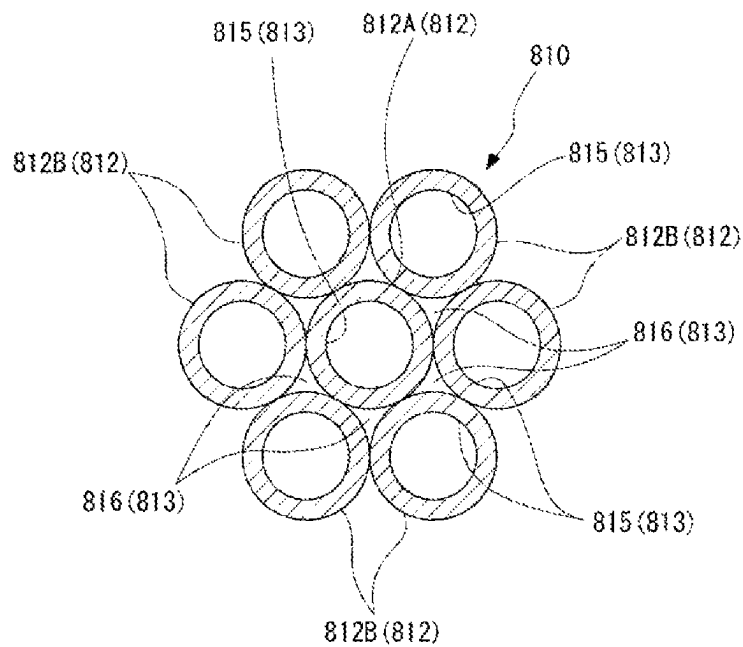


FIG. 13

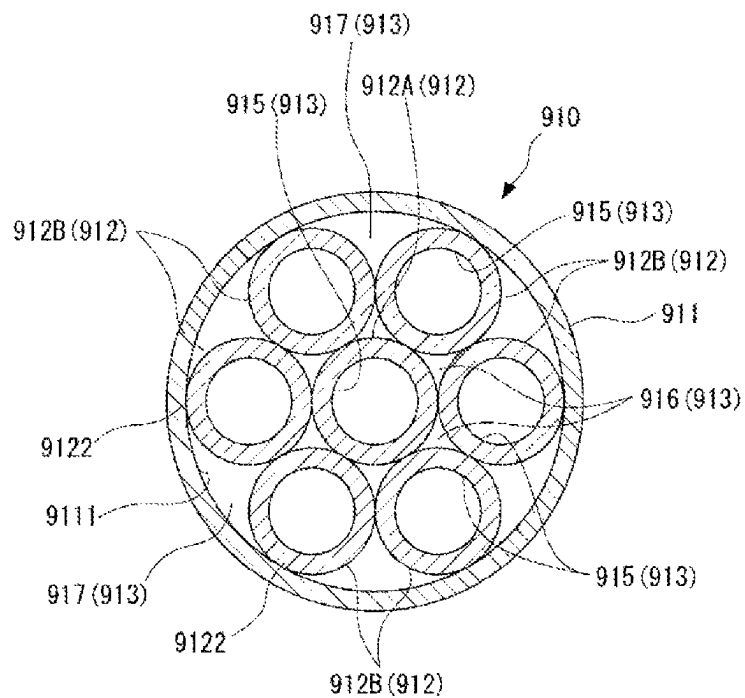


FIG. 14

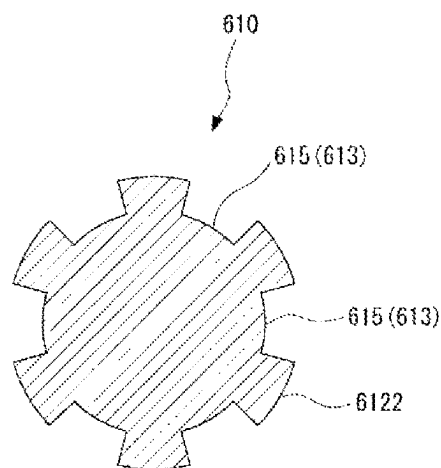
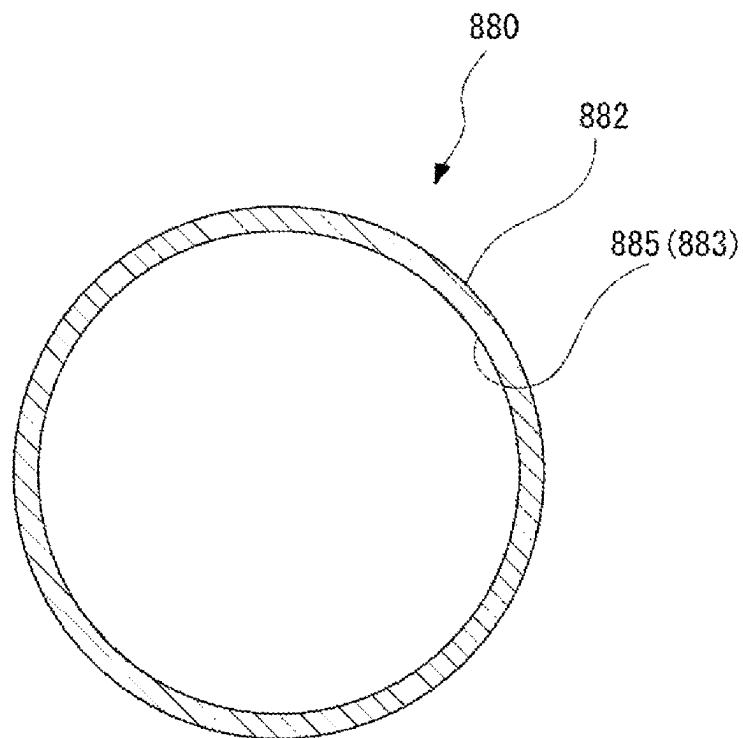


FIG. 15

**FIG. 16**

1

LIQUID COLLECTOR**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is the National Stage entry of International Application No. PCT/JP2021/020577, filed 30 May 2021, which claims priority to Japanese Application No. 2020-095795, filed 1 Jun. 2020.

BACKGROUND**Technical Field**

The present disclosure relates to a liquid collector.

Description of Related Art

A small amount of liquid may be collected by, for example, suction using a dropper or a syringe or absorption into test paper.

It is desirable to collect liquid efficiently. However, when, for example, a small amount of liquid is collected by using a dropper or the like, it may be difficult to efficiently collect all of the liquid.

Summary of Invention

Here, it is recognized, for example, without limitation, that efficient collection of liquid is desired.

SUMMARY

Some embodiments of the present disclosure provide a liquid collector. In some embodiments, the liquid collector collects liquid. In some embodiments, the liquid collector includes a flow passage member. In some embodiments, the flow passage member includes a flow passage that connects one end and other end of the flow passage member. In some embodiments, the liquid collector includes a contact member. In some embodiments, the contact member is disposed at the one end of the flow passage member. In some embodiments, the contact member includes an elastic material. In some embodiments, the contact member is configured to come into contact with a target from which liquid is to be collected. In some embodiments, the liquid collector includes a container. In some embodiments, the container receives the liquid discharged from the flow passage at the other end of the flow passage member.

Some embodiments of the present disclosure provide a liquid collector. In some embodiments, the liquid collector collects liquid. In some embodiments, the liquid collector includes a flow passage member. In some embodiments, the flow passage member includes a plurality of flow passages that connect one end and other end of the flow passage member. In some embodiments, the liquid collector includes a container. In some embodiments, the container receives the liquid discharged from the flow passages at the other end of the flow passage member.

According to the above-described embodiments, for example, the liquid can be efficiently collected.

Additional aspects and advantages of the present disclosure will become readily apparent to those skilled in this art from the following detailed description, wherein only illustrative embodiments of the present disclosure are shown and described. As will be realized, the present disclosure is capable of other and different embodiments, and its several

2

details are capable of modifications in various obvious respects, all without departing from the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a sectional view of a liquid collector according to an embodiment.

FIG. 2 is a sectional view of a flow passage member of the liquid collector illustrated in FIG. 1.

FIG. 3 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 1.

FIG. 4 is a sectional view of a liquid collector according to an embodiment.

FIG. 5 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 4.

FIG. 6 is a sectional view of a liquid collector according to an embodiment.

FIG. 7 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 6.

FIG. 8 is a sectional view of a liquid collector according to an embodiment.

FIG. 9 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 8.

FIG. 10 is a sectional view of a liquid collector according to an embodiment.

FIG. 11 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 10.

FIG. 12 is a sectional view of a flow passage member of a liquid collector according to an embodiment.

FIG. 13 is a sectional view of a flow passage member of a liquid collector according to an embodiment.

FIG. 14 is a sectional view of a flow passage member of a liquid collector according to an embodiment.

FIG. 15 is a sectional view of a flow passage member of a liquid collector according to an embodiment.

FIG. 16 is a sectional view of a flow passage member of a liquid collector according to an embodiment.

DESCRIPTION OF EMBODIMENTS

In some embodiments, a liquid collector collects liquid (sometimes also referred to as solution).

In some embodiments, the liquid may be body fluid produced by secretion in a target subject, or liquid other than body fluid. The liquid other than body fluid may be liquid adhering to a target object or liquid not adhering to a target object. The liquid not adhering to a target object may be liquid contained in a target object.

The liquid to be collected may be a solution. The liquid may be body fluid, a solution derived from body fluid, or diluted body fluid. The liquid may be a solution other than (not derived from) body fluid, or a mixture of body fluid or a solution derived from body fluid with a solution not derived from body fluid. The solution may be a solution used for sample measurement or a solution used for calibration measurement. For example, the solution may be a standard solution or a calibration solution. The sample to be measured may be a specimen.

Examples of the body fluid include lymph fluid, tissue fluid such as inter-tissue fluid, intercellular fluid, and interstitial fluid, coelomic fluid, serous cavity fluid, pleural fluid, ascites fluid, pericardial fluid, cerebrospinal fluid, synovial fluid, and aqueous humor. Examples of the body fluid further include digestive juice, such as saliva, gastric juice, bile, pancreatic juice, and intestinal juice, sweat, tear, nasal

mucus, urine, semen, vaginal fluid, amniotic fluid, and milk. The body fluid may be body fluid of an animal or a human. The “body fluid” may be a solution. The solution may include a physiological buffer, such as phosphate-buffered saline (PBS) or N-tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid (TES) buffer, containing a measurement target substance. The solution is not particularly limited as long as the measurement target substance is contained.

In some embodiments, the target subject may include, or may be, a human. In some embodiments, the target subject may include, or may be, a non-human animal. The non-human animal may include, or may be, a mammal. Examples of non-human animals include, but are not limited to, working animals, livestock animals, pet animals, and wild animals.

In some embodiments, the liquid collector includes a flow passage member. In some embodiments, the flow passage member may include a flow passage that connects one end and other end of the flow passage member. In some embodiments, the flow passage member may include a plurality of flow passages or only one flow passage. In some embodiments, when the flow passage member includes a plurality of flow passages, the flow passages extend in a pipe axis direction connecting the one end and the other end of the flow passage member. In some embodiments, the flow passages may or may not be substantially parallel to each other.

The term “pipe axis direction” used in this specification means a direction connecting the one end and the other end of the flow passage member.

In some embodiments, the flow passage member may have a columnar shape that extends in the pipe axis direction connecting the one end and the other end of the flow passage member. In some embodiments, the flow passage member having the columnar shape may have a plurality of grooves or only one groove that extends in the pipe axis direction along an outer peripheral surface of the flow passage member. In some embodiments, the groove or grooves each serves as a flow passage for the liquid. The cross-sectional shape of the groove or grooves may be, for example, without limitation, a U-shape (the corners may be substantially right angled, obtuse angled, acute angled, or rounded), a V-shape (the corner may be substantially acute angled or rounded), or a general trough shape. In some embodiments, the groove or grooves may extend straight in the pipe axis direction. In some embodiments, the groove or grooves may not extend straight in the pipe axis direction. For example, the groove or grooves may be formed in the flow passage member in a helical shape. For example, the groove or grooves may be inclined with respect to the pipe axis direction at the one end at which the liquid is absorbed.

In some embodiments, the flow passage member may include an outside tube. In some embodiments, the outside tube may have a tubular shape that extends in the pipe axis direction. In some embodiments, the flow passage member may include an inside member. In some embodiments, the inside member may be inserted in the outside tube. In some embodiments, the inside member may have a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the inside member. In some embodiments, an inner peripheral surface of the outside tube and the grooves in the inside member may define the flow passages. In some embodiments, the outside tube may cover at least a portion of the inside member in the pipe axis direction. In some

embodiments, the outside tube may cover a portion of the inside member in the pipe axis direction.

In some embodiments, the flow passage member may have a columnar shape that extends in the pipe axis direction. In some embodiments, the flow passage member having the columnar shape may have a circular shape or an elliptical shape in cross-section when viewed in the pipe axis direction. In some embodiments, the flow passage member may have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section. In some embodiments, the flow passage member may have a plurality of flow passages extending therethrough in the pipe axis direction to provide communication between the one end and the other end of the flow passage member. In some embodiments, the flow passage member may have a uniform cross-sectional shape in the pipe axis direction (for example, a cylindrical shape) or a cross-sectional shape that varies in the pipe axis direction (for example, decreases in size toward the one end).

In some embodiments, the flow passage member may be formed by bundling a plurality of single pipes that extend in the pipe axis direction. In some embodiments, each single pipe may have a tubular shape that extends in the pipe axis direction and include a flow passage that connects one end and the other end of the single pipe. In some embodiments, the single pipes may be joined together by adhesion or fusion bonding. In some embodiments, the single pipes may be bundled together with a band or the like. In some embodiments, the single pipes may be formed integrally with each other by using a resin material. When, for example, the single pipes are bundled together, holes in the single pipes and gaps between the single pipes serve as flow passages for the liquid.

In some embodiments, the flow passage member may include a porous material. The porous material may be, for example, without limitation, a porous ceramic or a sponge. The porous material may serve as a flow passage that allows fluid to flow therethrough by capillarity.

In some embodiments, the flow passage member may include an outer tube. In some embodiments, the outer tube may have a tubular shape that extends in the pipe axis direction. In some embodiments, the flow passage member may include a plurality of inner pipes. In some embodiments, the inner pipes may be inserted in the outer tube. In some embodiments, the inner pipes may extend in the pipe axis direction. When, for example, the inner pipes are disposed in the outer tube, holes formed in the inner pipes, gaps between the inner pipes, and gaps between an inner peripheral surface of the outer tube and the inner pipes disposed inside the inner peripheral surface serve as flow passages for fluid.

In some embodiments, the liquid collector includes a contact member. In some embodiments, the contact member may be disposed at the one end of the flow passage member. In some embodiments, the contact member may be configured to come into contact with a target from which the liquid is to be collected. In some embodiments, the contact member may include, or may be, an elastic material. In some embodiments, when the contact member includes an elastic material, the contact member may be a combination of a portion made of the elastic material and a portion made of another material. In some embodiments, the contact member may have mechanical, physical, chemical, and biological characteristics, such as elasticity and surface shape, such that the contact member can be brought into contact with, for example, a skin, a mucous membrane, or an eyeball of an

5

animal, such as a human, substantially without causing damage thereto. In some embodiments, the elastic material that forms the contact member may be softer than the flow passage member.

The term “elastic material” used in this specification generally refers to a material that is elastically deformable when the contact member is pressed against a target subject or a target object. The elastic material can be brought into contact with a target subject or a target object substantially without causing damage to a surface thereof in normal use. In some embodiments, the elastic material may include a material such as a silicone-based material, an urethane-based material (foam type, low-rebound material), a rubber material, a synthetic resin material (e.g., polyethylene), or a combination thereof (for example, without limitation, a structure including an inner portion made of low-rebound urethane and an outer portion made of silicone), or be formed mainly of a material that is one of, or a combination of, the above-mentioned materials.

In some embodiments, the contact member may include, or be made of, a hydrophilic material. In some embodiments, a contact surface may be hydrophilic. The contact member may have a hydrophilized surface. In some embodiments, the contact surface may be hydrophobic or water repellent. In some embodiments, the contact surface may be rough. In such a case, for example, a user (target subject) may recognize that the contact surface is in contact with their skin. In some embodiments, the contact surface may be embossed.

In some embodiments, the contact member may include an inlet opening. In some embodiments, the inlet opening may come into contact with a surface of the target from which the liquid is to be collected and allow introduction of the liquid that is on or near the surface. In some embodiments, the contact member may include an outlet opening. In some embodiments, the outlet opening may be connected to the inlet opening to provide fluid communication. In some embodiments, the outlet opening may be connected to the flow passage member to provide fluid communication. In some embodiments, the one end of the flow passage member may be fitted to the outlet opening in the contact member. In some embodiments, the contact member may include a sleeve. In some embodiments, the sleeve may have a tubular shape that extends toward the other end of the flow passage member. When, for example, the contact member includes the sleeve having a tubular shape that extends toward the other end of the flow passage member, the flow passage member is fitted to and covered with the sleeve. In some embodiments, the inlet opening and the outlet opening may have the same size or shape. In some embodiments, the inlet opening and the outlet opening may have different sizes or shapes. For example, the inlet opening may be relatively wide, and the thickness of a flow passage that provides fluid communication between the inlet opening and the outlet opening may be reduced toward the outlet opening. This arrangement may be reversed.

In some embodiments, the contact member may have a contact surface in a region around the inlet opening. In some embodiments, the contact surface may come into contact with a surface of the target from which the liquid is to be collected. In some embodiments, the contact surface may protrude from the one end of the flow passage member. In some embodiments, at least a portion of the contact surface may be spherical, or the entirety of the contact surface may be spherical.

In some embodiments, the liquid collector includes a container. In some embodiments, the container may receive

6

the liquid discharged from the flow passage or flow passages at the other end of the flow passage member. In some embodiments, the container may include a peripheral wall having a tubular shape. In some embodiments, the peripheral wall may extend in the pipe axis direction. In some embodiments, the container may include a bottom portion. In some embodiments, the bottom portion may block the peripheral wall at one side in the pipe axis direction. For example, the container may be formed of the peripheral wall and the bottom portion so as to have a tubular shape with a bottom. In some embodiments, the bottom portion of the container may be spaced from the other end of the flow passage member in the pipe axis direction. In some embodiments, a reservoir space that stores the liquid may be defined between the bottom portion of the container and the other end of the flow passage member. In some embodiments, the container may have a shape of a dish, a bowl, a tray, etc., having a recess capable of receiving the liquid.

In some embodiments, the container may have an opening at a side opposite to the side at which the bottom portion is provided in the pipe axis direction. In some embodiments, the contact member may be disposed to close the opening of the container. In some embodiments, the contact member may be mechanically connected to the opening of the container to close the opening of the container. In some embodiments, the contact member may be placed on the opening of the container to close the opening of the container without being mechanically connected to the opening of the container.

In some embodiments, the peripheral wall of the container may have an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction. In some embodiments, the other end of the flow passage member may be in contact with an inner peripheral surface of the peripheral wall of the container. When, for example, the other end of the flow passage member is in contact with the inner peripheral surface of the peripheral wall, the liquid in the flow passage or flow passages of the flow passage member easily moves to the inner peripheral surface of the peripheral wall.

In some embodiments, the container may or may not have an air hole that provides communication with the outside of the container. In some embodiments, the air hole may be a through hole connecting the inside and the outside of the container, or a gap between the container and the contact member. In the case where, for example, the container has the air hole, an amount of air corresponding to the volume of the liquid that flows into the container is discharged to the outside of the container through the air hole when the liquid flows into the container through the flow passage or flow passages. In the case where, for example, the container has no air hole, an amount of air corresponding to the volume of the liquid that flows into the container is discharged to the outside of the container through one or more of the flow passages when the liquid flows into the container through the flow passages.

In some embodiments, the air hole may be formed at a position such that the liquid that flows into the container does not flow out of the container when the liquid collector is used. In some embodiments, the air hole may be sealable. For example, a plug or seal that closes the air hole from the outside of the container may be provided. The air hole may be closed to seal the inside of the container. Accordingly, for example, without limitation, the liquid collected in the container can be prevented from leaking to the outside.

In some embodiments, the container may have a scale mark that indicates an amount of the liquid received in the

container. When, for example, the container has the scale mark, the amount of the liquid received in the container can be easily visually recognized. In some embodiments, the scale mark may be disposed to enable estimation of the amount of the liquid received in the container. In some

embodiments, the scale mark may be disposed at a position corresponding to a predetermined amount (required amount). Accordingly, for example, it can be determined whether a sufficient amount of liquid is collected. One, two, or more scale marks may be provided.

In some embodiments, the container may include a cone-shaped portion. In some embodiments, the cone-shaped portion may be disposed in the container. In some embodiments, at least a portion of the cone-shaped portion in the pipe axis direction may have an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction. In some embodiments, the cone-shaped portion may be formed over at least a partial region in the pipe axis direction, or over the entire region in the pipe axis direction. In some embodiments, the cone-shaped portion may be a component separate from the container or a part of the container. In some embodiments, when the cone-shaped portion is provided, the container may include no bottom portion and be open. In some embodiments, the other end of the flow passage member may be in contact with an inner peripheral surface of the cone-shaped portion. When, for example, when the other end of the flow passage member is in contact with the inner peripheral surface of the cone-shaped portion, the liquid in the flow passage or flow passages of the flow passage member easily moves to the inner peripheral surface of the cone-shaped portion. For example, when the other end of the flow passage member is in contact with the inner peripheral surface of the cone-shaped portion, movement of the flow passage member toward the bottom portion of the container is restricted, and the flow passage member is positioned with respect to the container in the pipe axis direction.

In some embodiments, the liquid collector may include a flow passage member and a container. In some embodiments, the flow passage member may include a plurality of flow passages that connect one end and other end of the flow passage member. In some embodiments, the container may receive liquid discharged from the flow passages at the other end of the flow passage member.

FIGS. 1 to 3 illustrate a liquid collector 100 according to an embodiment. As illustrated in FIGS. 1 to 3, the liquid collector 100 includes a flow passage member 110, a contact member 120, and a container 130. The liquid collector 100 collects liquid.

The flow passage member 110 includes an outside tube 111 and an inside member 112. The outside tube 111 has a tubular shape that extends in a pipe axis direction D_s connecting one end 1101 and other end 1102 of the flow passage member 110. In the embodiment of the present disclosure, the outside tube 111 has, for example, a circular shape in cross-section when viewed in the pipe axis direction D_s . The outside tube 111 may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction D_s . The outside tube 111 may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction D_s .

The inside member 112 is inserted in the outside tube 111. The inside member 112 has a columnar shape that extends in the pipe axis direction D_s . The inside member 112 has a uniform cross-sectional shape in the pipe axis direction D_s .

The inside member 112 has a plurality of grooves 1121 that extend in the pipe axis direction D_s . The grooves 1121 are arranged with equal intervals therebetween in a circumferential direction around a central axis of the inside member 112. Each groove 1121 is recessed radially inward from an outer peripheral surface 1122 of the inside member 112. Each groove 1121 extends continuously in the pipe axis direction D_s .

The outer peripheral surface 1122 of the inside member 112 inserted in the outside tube 111 is in contact with, or slightly spaced from, an inner peripheral surface 1111 of the outside tube 111. In the flow passage member 110, flow passages 113 are defined between the inner peripheral surface 1111 of the outside tube 111 and the grooves 1121 in the inside member 112. The flow passages 113 extend in the pipe axis direction D_s to connect the one end 1101 and the other end 1102 of the flow passage member 110.

Thus, the flow passage member 110 includes a plurality of flow passages 113. The flow passages 113 extend in the pipe axis direction D_s connecting the one end 1101 and the other end 1102. The flow passages 113 are substantially parallel to each other.

The contact member 120 is disposed at the one end 1101 of the flow passage member 110. The contact member 120 is configured to come into contact with a target from which the liquid is to be collected. The contact member 120 may include, or may be, an elastic material. When the contact member 120 includes an elastic material, the contact member 120 may be a combination of a portion made of the elastic material and a portion made of another material.

The contact member 120 has a flow-passage-member receiving hole 121. The flow-passage-member receiving hole 121 extends in the pipe axis direction D_s . The flow-passage-member receiving hole 121 has an inlet opening 1211 at an end thereof at one side in the pipe axis direction D_s . The inlet opening 1211 opens at a side opposite to a side at which the flow passage member 110 is disposed in the pipe axis direction D_s . The inlet opening 1211 comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface.

The flow-passage-member receiving hole 121 has an outlet opening 1212 at an end thereof at the other side in the pipe axis direction D_s . The outlet opening 1212 opens at the side at which the flow passage member 110 is disposed in the pipe axis direction D_s . The outlet opening 1212 is connected to the inlet opening 1211 to provide fluid communication through the flow-passage-member receiving hole 121.

The one end 1101 of the flow passage member 110 is fitted into the outlet opening 1212 of the contact member 120 from the other side in the pipe axis direction D_s . The one end 1101 of the flow passage member 110 is disposed at a position recessed from the inlet opening 1211 of the contact member 120 toward the other side in the pipe axis direction D_s . The one end 1101 of the flow passage member 110 does not project from the contact member 120 toward the one side in the pipe axis direction D_s . Thus, the outlet opening 1212 is connected to the flow passage member 110 to provide fluid communication.

The contact member 120 has a contact surface 123 in a region around the inlet opening 1211. The contact surface 123 extends over the entirety of a region of the contact member 120 excluding a region facing the container 130 at the other side in the pipe axis direction D_s . The contact surface 123 is configured to come into contact with the surface of the target from which the liquid is to be collected. The contact surface 123 projects from the one end 1101 of

the flow passage member **110** toward the one side in the pipe axis direction **Ds**. At least a portion of the contact surface **123** is spherical.

The container **130** receives the liquid discharged from the flow passages **113** at the other end **1102** of the flow passage member **110**. The container **130** includes a peripheral wall **131** and a bottom portion **132** that are integrated together.

The peripheral wall **131** has a tubular shape that extends in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the peripheral wall **131** has a uniform diameter in the pipe axis direction **Ds**. The peripheral wall **131** has an inner diameter greater than an outer diameter of the flow passage member **110**. Accordingly, an inner peripheral surface of the peripheral wall **131** and an outer peripheral surface of the flow passage member **110** are spaced from each other in a radial direction orthogonal to the pipe axis direction **Ds**.

The bottom portion **132** blocks the peripheral wall **131** at the other side in the pipe axis direction **Ds**. The bottom portion **132** is, for example, hemispherical. The bottom portion **132** may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction **Ds**. The container **130** is formed of the peripheral wall **131** and the bottom portion **132** so as to have a tubular shape with a bottom.

The container **130** has an opening **133** at the one side in the pipe axis direction **Ds** (side opposite to the side at which the bottom portion **132** is provided). The contact member **120** is disposed to close the opening **133** of the container **130**.

The bottom portion **132** of the container **130** is spaced from the other end **1102** of the flow passage member **110** in the pipe axis direction **Ds**. Accordingly, a reservoir space **138** that stores the liquid is defined between the bottom portion **132** of the container **130** and the other end **1102** of the flow passage member **110**.

At least a portion or the entirety of the container **130** may be made of a material having transparency characteristics that allow optical observation of the inside of the container **130** from the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container **130** from the outside include, but are not limited to, resin materials and glass materials. The material having transparency characteristics that allow optical observation of the inside of the container **130** from the outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container **130**.

The container **130** has scale marks **139** that indicate an amount of the liquid received in the container **130**.

To collect liquid in the above-described liquid collector **100**, the inlet opening **1211** of the contact member **120** is brought into contact with the surface of the target from which the liquid is to be collected. In some embodiments, the liquid collector **100** collects body fluid (liquid) secreted in a target subject. Examples of a part of the target subject in which the body fluid is secreted include, but are not limited to, an eyeball, regions around the eye at the inner and outer corners of the eye, a part in the oral cavity (for example, a sublingual region), a part in the nasal cavity, and the skin surface. In the embodiment of the present disclosure, the liquid collector **100** collects tear fluid from the outer corner the target subject's eye as the liquid. In this case, as illustrated in FIG. 3, the inlet opening **1211** of the contact member **120** is brought into contact with a surface **1000** of the target subject's face at or around the outer corner

of the eye. Accordingly, liquid **999** that is on or near the surface **1000** is introduced through the inlet opening **1211** by capillarity. Thus, the liquid **999** (tear fluid) on or near the surface **1000** is introduced through the inlet opening **1211**.

The liquid **999** introduced through the inlet opening **1211** enters the flow passages **113** at the one end **1101** of the flow passage member **110**, and flows through the flow passages **113** toward the other end **1102**. The liquid **999** is discharged from the flow passages **113** at the other end **1102** of the flow passage member **110**. The liquid **999** discharged from the flow passages **113** is stored in the reservoir space **138** of the container **130**. Thus, the liquid can be efficiently collected.

When the pipe axis direction **Ds** of the liquid collector **100** is inclined with respect to a vertical direction **Dv** upon collection of the liquid **999**, the liquid **999** introduced through the inlet opening **1211** flows downward in the vertical direction **Dv**. The liquid **999** introduced through the inlet opening **1211** tends to flow into a flow passage **113d**, which is one of the flow passages **113** in the flow passage member **110** that is positioned in a lower region in the vertical direction **Dv**. When the liquid **999** flows into the reservoir space **138**, an amount of air corresponding to the volume of the liquid **999** that flows into the reservoir space **138** is pushed out of the reservoir space **138**. The air pushed out of the reservoir space **138** moves toward the inlet opening **1211** through, for example, one or more of the flow passages **113** that are positioned in an upper region in the vertical direction **Dv** and through which the liquid **999** does not flow.

FIGS. 4 and 5 illustrate a liquid collector **200** according to an embodiment. As illustrated in FIGS. 4 and 5, the liquid collector **200** includes a flow passage member **210**, a contact member **220**, and a container **230**. The liquid collector **200** collects liquid.

The flow passage member **210** includes an outside tube **211** and an inside member **212**. The outside tube **211** has a tubular shape that extends in a pipe axis direction **Ds** connecting one end **2101** and other end **2102** of the flow passage member **210**. In the embodiment of the present disclosure, the outside tube **211** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **211** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **211** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction **Ds**.

The inside member **212** is inserted in the outside tube **211**. The inside member **212** has a columnar shape that extends in the pipe axis direction **Ds**. The inside member **212** has a uniform cross-sectional shape in the pipe axis direction **Ds**. The inside member **212** has a plurality of grooves **2121** that extend in the pipe axis direction **Ds**. The grooves **2121** are arranged with equal intervals therebetween in a circumferential direction around a central axis of the inside member **212**. Each groove **2121** is recessed radially inward from an outer peripheral surface **2122** of the inside member **212**. Each groove **2121** extends continuously in the pipe axis direction **Ds**.

The outer peripheral surface **2122** of the inside member **212** inserted in the outside tube **211** is in contact with, or slightly spaced from, an inner peripheral surface **2111** of the outside tube **211**. In the flow passage member **210**, flow passages **213** are defined between the inner peripheral surface **2111** of the outside tube **211** and the grooves **2121** in the inside member **212**. The flow passages **213** extend in

11

the pipe axis direction Ds to connect the one end **2101** and the other end **2102** of the flow passage member **210**.

Thus, the flow passage member **210** includes a plurality of flow passages **213**. The flow passages **213** extend in the pipe axis direction Ds connecting the one end **2101** and the other end **2102**. The flow passages **213** are substantially parallel to each other.

The contact member **220** is disposed at the one end **2101** of the flow passage member **210**. The contact member **220** is configured to come into contact with a target from which the liquid is to be collected. The contact member **220** may include, or may be, an elastic material. When the contact member **220** includes an elastic material, the contact member **220** may be a combination of a portion made of the elastic material and a portion made of another material.

The contact member **220** has a flow-passage-member receiving hole **221**. The flow-passage-member receiving hole **221** extends in the pipe axis direction Ds. The flow-passage-member receiving hole **221** has an inlet opening **2211** at an end thereof at one side in the pipe axis direction Ds. The inlet opening **2211** opens at a side opposite to a side at which the flow passage member **210** is disposed in the pipe axis direction Ds. The inlet opening **2211** comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface. The flow-passage-member receiving hole **221** has an outlet opening **2212** at an end thereof at the other side in the pipe axis direction Ds. The outlet opening **2212** opens at the side at which the flow passage member **210** is disposed in the pipe axis direction Ds. The outlet opening **2212** is connected to the inlet opening **2211** to provide fluid communication through the flow-passage-member receiving hole **221**.

The one end **2101** of the flow passage member **210** is fitted into the outlet opening **2212** of the contact member **220** from the other side in the pipe axis direction Ds. The one end **2101** of the flow passage member **210** is disposed at a position recessed from the inlet opening **2211** of the contact member **220** toward the other side in the pipe axis direction Ds. The one end **2101** of the flow passage member **210** does not project from the contact member **220** toward the one side in the pipe axis direction Ds. Thus, the outlet opening **2212** is connected to the flow passage member **210** to provide fluid communication.

The contact member **220** has a contact surface **223** in a region around the inlet opening **2211**. The contact surface **223** is configured to come into contact with the surface of the target from which the liquid is to be collected. The contact surface **223** projects from the one end **2101** of the flow passage member **210** toward the one side in the pipe axis direction Ds. At least a portion of the contact surface **223** is spherical. The contact surface **223** is spherical over the entirety of a region of the contact member **220** excluding a region facing the container **230** at the other side in the pipe axis direction Ds.

The container **230** receives the liquid discharged from the flow passages **213** at the other end **2102** of the flow passage member **210**. The container **230** includes a peripheral wall **231** and a bottom portion **232** that are integrated together.

The peripheral wall **231** has a tubular shape that extends in the pipe axis direction Ds. In the embodiment of the present disclosure, the peripheral wall **231** has a uniform diameter in the pipe axis direction Ds. The peripheral wall **231** has an inner diameter substantially equal to an outer diameter of the flow passage member **210**. Accordingly, the other end **2102** of the flow passage member **210** is in contact with an inner peripheral surface **2314** of the peripheral wall

12

231. When the other end **2102** of the flow passage member **210** is in contact with the inner peripheral surface **2314** of the peripheral wall **231** as described above, the liquid in the flow passages **213** of the flow passage member **210** easily moves along the inner peripheral surface **2314** of the peripheral wall **231**.

The bottom portion **232** blocks the peripheral wall **231** at the other side in the pipe axis direction Ds. The bottom portion **232** is, for example, hemispherical. The bottom portion **232** may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction Ds. The container **230** is formed of the peripheral wall **231** and the bottom portion **232** so as to have a tubular shape with a bottom.

The container **230** has an opening **233** at the one side in the pipe axis direction Ds (side opposite to the side at which the bottom portion **232** is provided). The contact member **220** is disposed to close the opening **233** of the container **230**.

The bottom portion **232** of the container **230** is spaced from the other end **2102** of the flow passage member **210** in the pipe axis direction Ds. Accordingly, a reservoir space **238** that stores the liquid is defined between the bottom portion **232** of the container **230** and the other end **2102** of the flow passage member **210**.

At least a portion or the entirety of the container **230** may be made of a material having transparency characteristics that allow optical observation of the inside of the container **230** from the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container **230** from the outside include, but are not limited to, resin materials and glass materials. The material having transparency characteristics that allow optical observation of the inside of the container **230** from the outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container **230**.

The container **230** has scale marks **239** that indicate an amount of the liquid received in the container **230**.

To collect liquid in the above-described liquid collector **200**, the inlet opening **2211** of the contact member **220** is brought into contact with a surface **1000** of the target from which the liquid is to be collected, as illustrated in FIG. 5. Accordingly, liquid **999** that is on or near the surface **1000** is introduced through the inlet opening **2211** by capillarity. The liquid **999** introduced through the inlet opening **2211** enters the flow passages **213** at the one end **2101** of the flow passage member **210**. The liquid **999** flows toward the other end **2102** through the flow passages **213**. The liquid **999** is discharged from the flow passages **213** at the other end **2102** of the flow passage member **210**. Since the outer peripheral surface **2122** of the flow passage member **210** is in contact with the inner peripheral surface **2314** of the container **230**, the liquid **999** discharged from the flow passages **213** easily flows along the inner peripheral surface **2314** of the container **230**. The liquid **999** flows downward along the inner peripheral surface **2314**, and is stored in the reservoir space **238** of the container **230**. Thus, the liquid can be efficiently collected.

When the pipe axis direction Ds of the liquid collector **200** is inclined with respect to a vertical direction Dv upon collection of the liquid **999**, the liquid **999** introduced through the inlet opening **2211** flows downward in the vertical direction Dv. The liquid **999** introduced through the inlet opening **2211** tends to flow into a flow passage **213d**,

13

which is one of the flow passages **213** in the flow passage member **210** that is positioned in a lower region in the vertical direction **Dv**.

When the liquid **999** flows into the reservoir space **238**, an amount of air corresponding to the volume of the liquid **999** that flows into the reservoir space **238** is pushed out of the reservoir space **238**. The air pushed out of the reservoir space **238** flows through one or more of the flow passages **213** that are positioned in an upper region in the vertical direction **Dv** and through which the liquid **999** does not flow, and is discharged to the outside from the inlet opening **2211**.

FIGS. **6** and **7** illustrate a liquid collector **300** according to an embodiment. As illustrated in FIGS. **6** and **7**, the liquid collector **300** includes a flow passage member **310**, a contact member **320**, a container **330**, and an inner member **340**. The liquid collector **300** collects liquid.

The flow passage member **310** includes an outside tube **311** and an inside member **312**.

The outside tube **311** has a tubular shape that extends in a pipe axis direction **Ds** connecting one end **3101** and other end **3102** of the flow passage member **310**. In the embodiment of the present disclosure, the outside tube **311** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **311** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **311** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction **Ds**.

The inside member **312** is inserted in the outside tube **311**. The inside member **312** has a columnar shape that extends in the pipe axis direction **Ds**. The inside member **312** has a uniform cross-sectional shape in the pipe axis direction **Ds**. The inside member **312** has a plurality of grooves **3121** that extend in the pipe axis direction **Ds**. The grooves **3121** are arranged with equal intervals therebetween in a circumferential direction around a central axis of the inside member **312**. Each groove **3121** is recessed radially inward from an outer peripheral surface **3122** of the inside member **312**. Each groove **3121** extends continuously in the pipe axis direction **Ds**.

The outer peripheral surface **3122** of the inside member **312** inserted in the outside tube **311** is in contact with, or slightly spaced from, an inner peripheral surface **3111** of the outside tube **311**. In the flow passage member **310**, flow passages **313** are defined between the inner peripheral surface **3111** of the outside tube **311** and the grooves **3121** in the inside member **312**. The flow passages **313** extend in the pipe axis direction **Ds** to connect the one end **3101** and the other end **3102** of the flow passage member **310**.

Thus, the flow passage member **310** includes a plurality of flow passages **313**. The flow passages **313** extend in the pipe axis direction **Ds** connecting the one end **3101** and the other end **3102**. The flow passages **313** are substantially parallel to each other.

The contact member **320** is disposed at the one end **3101** of the flow passage member **310**. The contact member **320** is configured to come into contact with a target from which the liquid is to be collected. The contact member **320** may include, or may be, an elastic material. When the contact member **320** includes an elastic material, the contact member **320** may be a combination of a portion made of the elastic material and a portion made of another material.

The contact member **320** has a flow-passage-member receiving hole **321**. The flow-passage-member receiving hole **321** extends in the pipe axis direction **Ds**. The flow-

14

passage-member receiving hole **321** has an inlet opening **3211** at an end thereof at one side in the pipe axis direction **Ds**. The inlet opening **3211** opens at a side opposite to a side at which the flow passage member **310** is disposed in the pipe axis direction **Ds**. The inlet opening **3211** comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface.

The flow-passage-member receiving hole **321** has an outlet opening **3212** at an end thereof at the other side in the pipe axis direction **Ds**. The outlet opening **3212** opens at the side at which the flow passage member **310** is disposed in the pipe axis direction **Ds**. The outlet opening **3212** is connected to the inlet opening **3211** to provide fluid communication through the flow-passage-member receiving hole **321**.

The one end **3101** of the flow passage member **310** is fitted into the outlet opening **3212** of the contact member **320** from the other side in the pipe axis direction **Ds**. The one end **3101** of the flow passage member **310** is disposed at a position recessed from the inlet opening **3211** of the contact member **320** toward the other side in the pipe axis direction **Ds**. The one end **3101** of the flow passage member **310** does not project from the contact member **320** toward the one side in the pipe axis direction **Ds**. Thus, the outlet opening **3212** is connected to the flow passage member **310** to provide fluid communication.

The contact member **320** has a contact surface **323** in a region around the inlet opening **3211**. The contact surface **323** is configured to come into contact with the surface of the target from which the liquid is to be collected. The contact surface **323** projects from the one end **3101** of the flow passage member **310** toward the one side in the pipe axis direction **Ds**. At least a portion of the contact surface **323** is spherical. The contact surface **323** is spherical over the entirety of a region of the contact member **320** excluding a region facing the container **330** at the other side in the pipe axis direction **Ds**.

The container **330** receives the liquid discharged from the flow passages **313** at the other end **3102** of the flow passage member **310**. The container **330** includes a peripheral wall **331** and a bottom portion **332** that are integrated together.

The peripheral wall **331** has a tubular shape that extends in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the peripheral wall **331** has an inner diameter that gradually decreases toward the bottom portion **332** at the other side from the one side in the pipe axis direction **Ds**.

The bottom portion **332** blocks the peripheral wall **331** at the other side in the pipe axis direction **Ds**. The bottom portion **332** is, for example, hemispherical. The bottom portion **332** may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction **Ds**. The container **330** is formed of the peripheral wall **331** and the bottom portion **332** so as to have a tubular shape with a bottom.

The container **330** has an opening **333** at the one side in the pipe axis direction **Ds** (side opposite to the side at which the bottom portion **332** is provided). The contact member **320** is disposed to close the opening **333** of the container **330**.

The bottom portion **332** of the container **330** is spaced from the other end **3102** of the flow passage member **310** in the pipe axis direction **Ds**. Accordingly, a reservoir space **338** that stores the liquid is defined between the bottom portion **332** of the container **330** and the other end **3102** of the flow passage member **310**.

15

At least a portion or the entirety of the container **330** may be made of a material having transparency characteristics that allow optical observation of the inside of the container **330** from the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container **330** from the outside include, but are not limited to, resin materials and glass materials. The material having transparency characteristics that allow optical observation of the inside of the container **330** from the outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container **330**.

The container **330** has scale marks **339** that indicate an amount of the liquid received in the container **330**.

The inner member **340** is disposed in the container **330**. The inner member **340** includes a cone-shaped portion **341** having an inner diameter that gradually decreases toward the bottom portion **332** in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the cone-shaped portion **341** is formed over the entire region of the inner member **340** in the pipe axis direction **Ds**. In some embodiments, the cone-shaped portion **341** may be formed over only a partial region of the inner member **340** in the pipe axis direction **Ds**. The cone-shaped portion **341** has a first opening **3411** at an end adjacent to the opening **333**, and an inner diameter of the first opening **3411** is greater than an outer diameter of the flow passage member **310** at the other end **3102**. The cone-shaped portion **341** has a second opening **3412** at an end adjacent to the bottom portion **332**, and an inner diameter of the second opening **3412** is less than the outer diameter of the flow passage member **310** at the other end **3102**. Accordingly, the other end **3102** of the flow passage member **310** is in contact with an inner peripheral surface of the cone-shaped portion **341** at a position between the first opening **3411** and the second opening **3412** of the cone-shaped portion **341**. When the other end **3102** of the flow passage member **310** is in contact with the cone-shaped portion **341** as described above, the liquid in the flow passages **313** of the flow passage member **310** easily flows along the inner peripheral surface of the cone-shaped portion **341**.

To collect liquid in the above-described liquid collector **300**, the inlet opening **3211** of the contact member **320** is brought into contact with a surface **1000** of the target from which the liquid is to be collected, as illustrated in FIG. 7. Accordingly, liquid **999** that is on or near the surface **1000** is introduced through the inlet opening **3211** by capillarity. The liquid **999** introduced through the inlet opening **3211** enters the flow passages **313** at the one end **3101** of the flow passage member **310**. The liquid **999** flows toward the other end **3102** through the flow passages **313**. The liquid **999** is discharged from the flow passages **313** at the other end **3102** of the flow passage member **310**. The liquid **999** discharged from the flow passages **313** flows downward along the inner peripheral surface of the cone-shaped portion **341**. The liquid **999** falls from the second opening **3412** of the cone-shaped portion **341**, and is stored in the reservoir space **338** of the container **330**. Thus, the liquid can be efficiently collected.

FIGS. **8** and **9** illustrate a liquid collector **400** according to an embodiment. As illustrated in FIGS. **8** and **9**, the liquid collector **400** includes a flow passage member **410**, a contact member **420**, a container **430**, and an inner member **440**. The liquid collector **400** collects liquid.

The flow passage member **410** includes an outside tube **411** and an inside member **412**.

16

The outside tube **411** has a tubular shape that extends in a pipe axis direction **Ds** connecting one end **4101** and other end **4102** of the flow passage member **410**. In the embodiment of the present disclosure, the outside tube **411** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **411** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **411** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction **Ds**.

The inside member **412** is inserted in the outside tube **411**. The inside member **412** has a columnar shape that extends in the pipe axis direction **Ds**. The inside member **412** has a uniform cross-sectional shape in the pipe axis direction **Ds**. The inside member **412** has a plurality of grooves **4121** that extend in the pipe axis direction **Ds**. The grooves **4121** are arranged with equal intervals therebetween in a circumferential direction around a central axis of the inside member **412**. Each groove **4121** is recessed radially inward from an outer peripheral surface **4122** of the inside member **412**. Each groove **4121** extends continuously in the pipe axis direction **Ds**.

The outer peripheral surface **4122** of the inside member **412** inserted in the outside tube **411** is in contact with, or slightly spaced from, an inner peripheral surface **4111** of the outside tube **411**. In the flow passage member **410**, flow passages **413** are defined between the inner peripheral surface **4111** of the outside tube **411** and the grooves **4121** in the inside member **412**. The flow passages **413** extend in the pipe axis direction **Ds** to connect the one end **4101** and the other end **4102** of the flow passage member **410**.

Thus, the flow passage member **410** includes a plurality of flow passages **413**. The flow passages **413** extend in the pipe axis direction **Ds** connecting the one end **4101** and the other end **4102**. The flow passages **413** are substantially parallel to each other.

The contact member **420** is disposed at the one end **4101** of the flow passage member **410**. The contact member **420** is configured to come into contact with a target from which the liquid is to be collected. The contact member **420** may include, or may be, an elastic material. When the contact member **420** includes an elastic material, the contact member **420** may be a combination of a portion made of the elastic material and a portion made of another material.

The contact member **420** has a flow-passage-member receiving hole **421**. The flow-passage-member receiving hole **421** extends in the pipe axis direction **Ds**. The flow-passage-member receiving hole **421** has an inlet opening **4211** at an end thereof at one side in the pipe axis direction **Ds**. The inlet opening **4211** opens at a side opposite to a side at which the flow passage member **410** is disposed in the pipe axis direction **Ds**. The inlet opening **4211** comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface.

The contact member **420** includes a sleeve **425** that is integral therewith. The sleeve **425** projects from the contact member **420** toward the other end **4102** of the flow passage member **410**. The sleeve **425** extends in the pipe axis direction **Ds**. The sleeve **425** is inserted in the container **430**.

The flow-passage-member receiving hole **421** has an outlet opening **4212** at an end thereof at the other side in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the outlet opening **4212** is formed in the sleeve **425**. The outlet opening **4212** opens at the side at which the

17

flow passage member 410 is disposed in the pipe axis direction Ds. The outlet opening 4212 is connected to the inlet opening 4211 to provide fluid communication through the flow-passage-member receiving hole 421.

The one end 4101 of the flow passage member 410 is fitted into the outlet opening 4212 of the contact member 420 from the other side in the pipe axis direction Ds. Since the contact member 420 includes the sleeve 425, the one end 4101 of the flow passage member 410 is retained by the contact member 420 (including the sleeve 425) over a longer length in the pipe axis direction Ds. Accordingly, the flow passage member 410 is more strongly supported. Thus, the outlet opening 4212 is connected to the flow passage member 410 to provide fluid communication.

The one end 4101 of the flow passage member 410 is disposed at a position recessed from the inlet opening 4211 of the contact member 420 toward the other side in the pipe axis direction Ds. The one end 4101 of the flow passage member 410 does not project from the contact member 420 toward the one side in the pipe axis direction Ds.

The contact member 420 has a contact surface 423 in a region around the inlet opening 4211. The contact surface 423 is configured to come into contact with the surface of the target from which the liquid is to be collected. The contact surface 423 projects from the one end 4101 of the flow passage member 410 toward the one side in the pipe axis direction Ds. At least a portion of the contact surface 423 is spherical. The contact surface 423 is spherical over the entirety of a region of the contact member 420 excluding a region facing the container 430 at the other side in the pipe axis direction Ds.

The container 430 receives the liquid discharged from the flow passages 413 at the other end 4102 of the flow passage member 410. The container 430 includes a peripheral wall 431 and a bottom portion 432 that are integrated together.

The peripheral wall 431 has a tubular shape that extends in the pipe axis direction Ds. In the embodiment of the present disclosure, the peripheral wall 431 has an inner diameter that gradually decreases toward the bottom portion 432 at the other side from the one side in the pipe axis direction Ds.

The bottom portion 432 blocks the peripheral wall 431 at the other side in the pipe axis direction Ds. The bottom portion 432 is, for example, hemispherical. The bottom portion 432 may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction Ds. The container 430 is formed of the peripheral wall 431 and the bottom portion 432 so as to have a tubular shape with a bottom.

The container 430 has an opening 433 at the one side in the pipe axis direction Ds (side opposite to the side at which the bottom portion 432 is provided). The contact member 420 is disposed to close the opening 433 of the container 430.

The bottom portion 432 of the container 430 is spaced from the other end 4102 of the flow passage member 410 in the pipe axis direction Ds. Accordingly, a reservoir space 438 that stores the liquid is defined between the bottom portion 432 of the container 430 and the other end 4102 of the flow passage member 410.

At least a portion or the entirety of the container 430 may be made of a material having transparency characteristics that allow optical observation of the inside of the container 430 from the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container 430 from the outside include, but are not limited to, resin materials and glass materials. The

18

material having transparency characteristics that allow optical observation of the inside of the container 430 from the outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container 430.

The container 430 has scale marks 439 that indicate an amount of the liquid received in the container 430.

The inner member 440 is disposed in the container 430.

The inner member 440 includes a cone-shaped portion 441 having an inner diameter that gradually decreases toward the bottom portion 432 in the pipe axis direction Ds. In the embodiment of the present disclosure, the cone-shaped portion 441 is formed over the entire region of the inner member 440 in the pipe axis direction Ds. In some embodiments, the cone-shaped portion 441 may be formed over only a partial region of the inner member 440 in the pipe axis direction Ds. The cone-shaped portion 441 has a first opening 4411 at an end adjacent to the opening 433, and an inner diameter of the first opening 4411 is greater than an outer diameter of the flow passage member 410 at the other end 4102. The cone-shaped portion 441 has a second opening 4412 at an end adjacent to the bottom portion 432, and an inner diameter of the second opening 4412 is less than the outer diameter of the flow passage member 410 at the other end 4102. Accordingly, the other end 4102 of the flow passage member 410 is in contact with an inner peripheral surface of the cone-shaped portion 441 at a position between the first opening 4411 and the second opening 4412 of the cone-shaped portion 441. When the other end 4102 of the flow passage member 410 is in contact with the cone-shaped portion 441 as described above, the liquid in the flow passages 413 of the flow passage member 410 easily flows along the cone-shaped portion 441.

To collect liquid in the above-described liquid collector 400, the inlet opening 4211 of the contact member 420 is brought into contact with a surface 1000 of the target from which the liquid is to be collected, as illustrated in FIG. 9. Accordingly, liquid 999 that is on or near the surface 1000 is introduced through the inlet opening 4211 by capillarity. The liquid 999 introduced through the inlet opening 4211 enters the flow passages 413 at the one end 4101 of the flow passage member 410. The liquid 999 flows toward the other end 4102 through the flow passages 413. The liquid 999 is discharged from the flow passages 413 at the other end 4102 of the flow passage member 410. The liquid 999 discharged from the flow passages 413 flows downward along the inner peripheral surface of the cone-shaped portion 441. The liquid 999 falls from the second opening 4412 of the cone-shaped portion 441, and is stored in the reservoir space 438 of the container 430. Thus, the liquid can be efficiently collected.

FIGS. 10 and 11 illustrate a liquid collector 500 according to an embodiment. As illustrated in FIGS. 10 and 11, the liquid collector 500 includes a flow passage member 510, a container 530, and an inner member 540. The liquid collector 500 collects liquid.

The flow passage member 510 includes an outside tube 511 and an inside member 512.

The outside tube 511 has a tubular shape that extends in a pipe axis direction Ds connecting one end 5101 and other end 5102 of the flow passage member 510. In the embodiment of the present disclosure, the outside tube 511 has, for example, a circular shape in cross-section when viewed in the pipe axis direction Ds. The outside tube 511 may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction Ds. The outside tube 511

may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction Ds.

The inside member **512** is inserted in the outside tube **511**. The inside member **512** has a columnar shape that extends in the pipe axis direction Ds. The inside member **512** has a uniform cross-sectional shape in the pipe axis direction Ds. The inside member **512** has a plurality of grooves **5121** that extend in the pipe axis direction Ds. The grooves **5121** are arranged with equal intervals therebetween in a circumferential direction around a central axis of the inside member **512**. Each groove **5121** is recessed radially inward from an outer peripheral surface **5122** of the inside member **512**. Each groove **5121** extends continuously in the pipe axis direction Ds.

The outer peripheral surface **5122** of the inside member **512** inserted in the outside tube **511** is in contact with, or slightly spaced from, an inner peripheral surface **5111** of the outside tube **511**. In the flow passage member **510**, flow passages **513** are defined between the inner peripheral surface **5111** of the outside tube **511** and the grooves **5121** in the inside member **512**. The flow passages **513** extend in the pipe axis direction Ds to connect the one end **5101** and the other end **5102** of the flow passage member **510**.

Thus, the flow passage member **510** includes a plurality of flow passages **513**. The flow passages **513** extend in the pipe axis direction Ds connecting the one end **5101** and the other end **5102**. The flow passages **513** are substantially parallel to each other.

The container **530** receives the liquid discharged from the flow passages **513** at the other end **5102** of the flow passage member **510**. The container **530** includes a peripheral wall **531** and a bottom portion **532** that are integrated together.

The peripheral wall **531** has a tubular shape that extends in the pipe axis direction Ds. In the embodiment of the present disclosure, the peripheral wall **531** has an inner diameter that gradually decreases toward the bottom portion **532** at the other side from the one side in the pipe axis direction Ds.

The bottom portion **532** blocks the peripheral wall **531** at the other side in the pipe axis direction Ds. The bottom portion **532** is, for example, hemispherical. The bottom portion **532** may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction Ds. The container **530** is formed of the peripheral wall **531** and the bottom portion **532** so as to have a tubular shape with a bottom.

The container **530** has an opening **533** at one side in the pipe axis direction Ds (side opposite to the side at which the bottom portion **532** is provided). The flow passage member **510** is inserted into the container **530** through the opening **533**. The bottom portion **532** of the container **530** is spaced from the other end **5102** of the flow passage member **510** in the pipe axis direction Ds. Accordingly, a reservoir space **538** that stores the liquid is defined between the bottom portion **532** of the container **530** and the other end **5102** of the flow passage member **510**.

At least a portion or the entirety of the container **530** may be made of a material having transparency characteristics that allow optical observation of the inside of the container **530** from the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container **530** from the outside include, but are not limited to, resin materials and glass materials. The material having transparency characteristics that allow optical observation of the inside of the container **530** from the

outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container **530**.

The container **530** has scale marks **539** that indicate an amount of the liquid received in the container **530**.

The inner member **540** is disposed in the container **530**. The inner member **540** includes a cone-shaped portion **541** having an inner diameter that gradually decreases toward the bottom portion **532** in the pipe axis direction Ds. In the embodiment of the present disclosure, the cone-shaped portion **541** is formed over the entire region of the inner member **540** in the pipe axis direction Ds. In some embodiments, the cone-shaped portion **541** may be formed over only a partial region of the inner member **540** in the pipe axis direction Ds. The cone-shaped portion **541** has a first opening **5411** at an end adjacent to the opening **533**, and an inner diameter of the first opening **5411** is greater than an outer diameter of the flow passage member **510** at the other end **5102**. The cone-shaped portion **541** has a second opening **5412** at an end adjacent to the bottom portion **532**, and an inner diameter of the second opening **5412** is less than the outer diameter of the flow passage member **510** at the other end **5102**. Accordingly, the other end **5102** of the flow passage member **510** is in contact with an inner peripheral surface of the cone-shaped portion **541** at a position between the first opening **5411** and the second opening **5412** of the cone-shaped portion **541**. When the other end **5102** of the flow passage member **510** is in contact with the cone-shaped portion **541** as described above, the liquid in the flow passages **513** of the flow passage member **510** easily flows along the cone-shaped portion **541**.

To collect liquid in the above-described liquid collector **500**, the one end **5101** of the flow passage member **510** is brought into contact with a surface **1000** of the target from which the liquid is to be collected, as illustrated in FIG. **11**. The one end **5101** may have a hydrophilic or embossed surface. Accordingly, liquid **999** that is on or near the surface **1000** is introduced from the one end **5101** of the flow passage member **510** by capillarity. The liquid **999** introduced into the flow passage member **510** flows toward the other end **5102** through the flow passages **513**. The liquid **999** is discharged from the flow passages **513** at the other end **5102** of the flow passage member **510**. The liquid **999** discharged from the flow passages **513** flows downward along the inner peripheral surface of the cone-shaped portion **541**. The liquid **999** falls from the second opening **5412** of the cone-shaped portion **541**, and is stored in the reservoir space **538** of the container **530**. Thus, the liquid can be efficiently collected.

FIG. **12** illustrates a flow passage member **710** of a liquid collector according to an embodiment. Referring to FIG. **12**, the flow passage member **710** has a columnar shape that extends in a pipe axis direction Ds connecting one end and other end of the flow passage member **710** (direction orthogonal to the plane of FIG. **12**). In the embodiment of the present disclosure, the flow passage member **710** has, for example, a circular shape in cross-section when viewed in the pipe axis direction Ds. The flow passage member **710** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction Ds. The flow passage member **710** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction Ds.

21

The flow passage member **710** has a plurality of through holes **715** that extend therethrough in the pipe axis direction **Ds** to connect one end and the other end of the flow passage member **710**. In the embodiment of the present disclosure, a through hole **715A**, which is one of the through holes **715**, is disposed at the center of the flow passage member **710**. Through holes **715B**, which are the remaining ones of the through holes **715**, are arranged in an outer peripheral region of the flow passage member **710** with equal intervals therebetween in the circumferential direction. The through holes **715** are substantially parallel to the pipe axis direction **Ds**.

Each through hole **715** serves as a flow passage **713** for the liquid in the flow passage member **710**. The flow passage member **710** includes a plurality of flow passages **713** (through holes **715**). The flow passages **713** extend in the pipe axis direction **Ds** connecting the one end and the other end. The flow passages **713** are substantially parallel to each other.

In some embodiments, the above-described flow passage member **710** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

FIG. **13** illustrates a flow passage member **810** of a liquid collector according to an embodiment. As illustrated in FIG. **13**, the flow passage member **810** includes a plurality of single pipes **812**.

Each single pipe **812** has a tubular shape that extends in a pipe axis direction **Ds** (direction orthogonal to the plane of FIG. **13**). Each single pipe **812** has a uniform cross-sectional shape in the pipe axis direction **Ds**. In the embodiment of the present disclosure, each single pipe **812** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The flow passage member **810** is formed by bundling the single pipes **812**. The single pipes **812** may be bundled and joined together by, for example, without limitation, adhesion, fusion bonding, or welding. The single pipes **812** may be bundled together by using, for example, without limitation, a belt-shaped band or a ring. In some embodiments, each single pipe may have a cross-sectional shape that varies in the pipe axis direction. For example, each single pipe may be a combination of two single pipes having different sizes (for example, hole diameters).

A single pipe **812A**, which is one of the single pipes **812**, is disposed at the center of the flow passage member **810**. Single pipes **812B**, which are the remaining ones of the single pipes **812**, are arranged in contact with each other in the circumferential direction in a region radially outside the single pipe **812A**. Each single pipe **812** extends continuously in the pipe axis direction **Ds**. Each single pipe **812** has a hole **815** that extends therethrough in the pipe axis direction **Ds**.

A plurality of gaps **816** are formed between the single pipe **812A** and the single pipes **812B** arranged radially outside the single pipe **812A**. Each gap **816** extends continuously in the pipe axis direction **Ds** to connect one end and the other end of the flow passage member **810**.

Thus, the holes **815** and the gaps **816** are defined in the flow passage member **810**. The holes **815** and the gaps **816** each serve as a flow passage **813** for the liquid in the flow passage member **810**. The flow passage member **810** has a plurality of flow passages **813** (holes **815** and gaps **816**). The flow passages **813** extend in the pipe axis direction **Ds** connecting the one end and the other end of the flow passage member **810**. The flow passages **813** are substantially parallel to each other.

In some embodiments, the above-described flow passage member **810** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

22

FIG. **14** illustrates a flow passage member **910** of a liquid collector according to an embodiment. Referring to FIG. **14**, the flow passage member **910** includes an outer tube **911** and inner pipes **912**.

The outer tube **911** has a tubular shape that extends in a pipe axis direction **Ds** connecting one end and other end of the flow passage member **910** (direction orthogonal to the plane of FIG. **14**). In the embodiment of the present disclosure, the outer tube **911** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**.

The inner pipes **912** are inserted in the outer tube **911**. Each inner pipe **912** has a tubular shape that extends in the pipe axis direction **Ds**. Each inner pipe **912** has a uniform cross-sectional shape in the pipe axis direction **Ds**. In the embodiment of the present disclosure, each inner pipe **912** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The inner pipes **912** have an outer diameter less than an inner diameter of the outer tube **911**. The inner pipes **912** are inserted in the outer tube **911**. An inner pipe **912A**, which is one of the inner pipes **912**, is disposed at the center of the outer tube **911**. Inner pipes **912B**, which are the remaining ones of the inner pipes **912**, are arranged in contact with each other in the circumferential direction in a region radially outside the inner pipe **912A**. Each inner pipe **912** extends continuously in the pipe axis direction **Ds**. Each inner pipe **912** has a hole **915** that extends therethrough in the pipe axis direction **Ds**. In some embodiments, each single pipe may have a cross-sectional shape that varies in the pipe axis direction. For example, each single pipe may be a combination of two single pipes having different sizes (for example, hole diameters).

A plurality of gaps **916** are formed between the inner pipe **912A** and the inner pipes **912B** arranged radially outside the inner pipe **912A**. Each gap **916** extends continuously in the pipe axis direction **Ds** to connect the one end and the other end of the flow passage member **910**.

The inner pipes **912B** inserted in the outer tube **911** have outer peripheral surfaces **9122** that are in contact with, or slightly spaced from, an inner peripheral surface **9111** of the outer tube **911**. In the flow passage member **910**, the inner peripheral surface **9111** of the outer tube **911** and the outer peripheral surfaces **9122** of two of the inner pipes **912B** that are adjacent to each other in the circumferential direction define an outer peripheral gap **917** therebetween. The outer peripheral gap **917** continuously extends in the pipe axis direction **Ds** to connect one end and the other end of the flow passage member **910**.

Thus, the holes **915**, the gaps **916**, and the outer peripheral gaps **917** are defined in the flow passage member **910**. The holes **915**, the gaps **916**, and the outer peripheral gaps **917** each serve as a flow passage **913** for the liquid in the flow passage member **910**. The flow passage member **910** has a plurality of flow passages **913** (holes **915**, gaps **916**, and outer peripheral gaps **917**). The flow passages **913** extend in the pipe axis direction **Ds** connecting the one end and the other end of the flow passage member **910**. The flow passages **913** are substantially parallel to each other.

In some embodiments, the above-described flow passage member **910** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

FIG. **15** illustrates a flow passage member **610** of a liquid collector according to an embodiment. Referring to FIG. **15**, the flow passage member **610** has a columnar shape that extends in a pipe axis direction **Ds** connecting one end and other end of the flow passage member **610** (direction orthogonal to the plane of FIG. **15**). The flow passage member **610** has a uniform cross-sectional shape in the pipe

axis direction Ds. The flow passage member **610** has a plurality of grooves **615**. The grooves **615** are arranged with equal intervals therebetween in a circumferential direction around a central axis of the flow passage member **610**. Each groove **615** is recessed radially inward from an outer peripheral surface **6122** of the flow passage member **610**. Each groove **615** extends continuously in the pipe axis direction Ds.

Each groove **615** serves as a flow passage **613** for the liquid in the flow passage member **610**. The flow passage member **610** includes a plurality of flow passages **613** (grooves **615**). The flow passages **613** extend in the pipe axis direction Ds connecting the one end and the other end of the flow passage member **610**. The flow passages **613** are substantially parallel to each other.

In some embodiments, the above-described flow passage member **610** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

FIG. **16** illustrates a flow passage member **880** of a liquid collector according to an embodiment. As illustrated in FIG. **16**, the flow passage member **880** includes one pipe **882**.

The pipe **882** has a tubular shape that extends in a pipe axis direction Ds (direction orthogonal to the plane of FIG. **16**). The pipe **882** has a uniform cross-sectional shape in the pipe axis direction Ds. In the embodiment of the present disclosure, the pipe **882** has, for example, a circular shape in cross-section when viewed in the pipe axis direction Ds. The pipe **882** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction Ds. The pipe **882** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction Ds. The pipe **882** has a hole **885** that extends therethrough in the pipe axis direction Ds.

The hole **885** serves as a flow passage **883** for the liquid in the flow passage member **880**. The flow passage member **880** includes the flow passage **883** (hole **885**). The flow passage **883** extends in the pipe axis direction Ds connecting one end and other end of the flow passage member **880**.

In some embodiments, the above-described flow passage member **880** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

The present disclosure also provides the following embodiments:

A001

A liquid collector that collects liquid, the liquid collector including:

- a flow passage member including a flow passage that connects one end and other end of the flow passage member;
- a contact member that is disposed at the one end of the flow passage member, that includes an elastic material, and that is configured to come into contact with a target from which liquid is to be collected; and
- a container that receives the liquid discharged from the flow passage at the other end of the flow passage member.

A101

The liquid collector according to Embodiment A001, wherein the flow passage member includes a plurality of the flow passages that extend in a pipe axis direction connecting the one end and the other end of the flow passage member.

A102

The liquid collector according to Embodiment A101, wherein the plurality of the flow passages are substantially parallel to each other.

5 A103

The liquid collector according to Embodiment A101 or A102, wherein the flow passage member has a columnar shape that extends in the pipe axis direction connecting the one end and the other end of the flow passage member, and has a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the flow passage member.

10 A104

The liquid collector according to Embodiment A101 or A102, wherein the flow passage member includes an outside tube having a tubular shape that extends in the pipe axis direction, and an inside member inserted in the outside tube and having a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the flow passage member, and wherein an inner peripheral surface of the outside tube and the plurality of grooves in the inside member define the plurality of the flow passages.

15 A105

The liquid collector according to Embodiment A101 or A102, wherein the flow passage member has a columnar shape that extends in the pipe axis direction, and the plurality of the flow passages extend through the flow passage member in the pipe axis direction to provide communication between the one end and the other end of the flow passage member.

20 A106

The liquid collector according to Embodiment A101 or A102, wherein the flow passage member includes a plurality of single pipes that are bundled together and that extend in the pipe axis direction.

25 A107

The liquid collector according to Embodiment A101 or A102, wherein the flow passage member includes an outer tube having a tubular shape that extends in the pipe axis direction, and a plurality of inner pipes that are inserted in the outer tube and that extend in the pipe axis direction.

30 A111

The liquid collector according to any one of Embodiments A001 to A107, wherein the container includes a peripheral wall having a tubular shape that extends in the pipe axis direction, and a bottom portion that blocks the peripheral wall at one side in the pipe axis direction.

35 A112

The liquid collector according to Embodiment A111, wherein the bottom portion of the container is spaced from the other end of the flow passage member in the pipe axis direction, and wherein a reservoir space that stores the liquid is defined between the bottom portion of the container and the other end of the flow passage member.

60

65

A113

The liquid collector according to Embodiment A111 or A112, wherein the peripheral wall of the container has an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction.

A114

The liquid collector according to any one of Embodiments A111 to A113, wherein the other end of the flow passage member is in contact with an inner peripheral surface of the peripheral wall of the container.

A115

The liquid collector according to any one of Embodiments A111 to A114, wherein the container has an air hole that provides communication with an outside of the container.

A116

The liquid collector according to any one of Embodiments A111 to A115, wherein the container has a scale mark that indicates an amount of the liquid received in the

A121

The liquid collector according to any one of Embodiments A101 to A115, further including:

a cone-shaped portion disposed in the container, wherein at least a portion of the cone-shaped portion in the pipe axis direction has an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction.

A122

The liquid collector according to Embodiment A121, wherein the other end of the flow passage member is in contact with an inner peripheral surface of the cone-shaped portion.

A131

The liquid collector according to any one of Embodiments A001 to A122,

wherein the contact member includes

an inlet opening that comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface, and

an outlet opening that is connected to the inlet opening to provide fluid communication and also connected to the flow passage member to provide fluid communication.

A132

The liquid collector according to Embodiment A131, wherein the one end of the flow passage member is fitted to the outlet opening of the contact member.

A133

The liquid collector according to Embodiment A132, wherein the contact member includes a sleeve that has a tubular shape and extends toward the other end of the flow passage member.

A134

The liquid collector according to any one of Embodiments A131 to A133, wherein the contact member has a contact surface in a region around the inlet opening, the contact surface coming into contact with the surface of the target from which the liquid is to be collected.

A135

The liquid collector according to Embodiment A134, wherein at least a portion of the contact surface is spherical.

A136

The liquid collector according to any one of Embodiments A101 to A135,

wherein the container has an opening at a side opposite to a side at which the bottom portion is provided in the pipe axis direction, and wherein the contact member is disposed to close the opening.

A002

A liquid collector that collects liquid, the liquid collector including:

a flow passage member including a plurality of flow passages that connect one end and other end of the flow passage member; and

a container that receives the liquid discharged from the plurality of flow passages at the other end of the flow passage member.

A201

The liquid collector according to Embodiment A002, further including:

a cone-shaped portion disposed in the container, wherein at least a portion of the cone-shaped portion in the pipe axis direction has an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction.

While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. It is not intended that the invention be limited by the specific examples provided within the specification. While the invention has been described with reference to the aforementioned specification, the descriptions and illustrations of the embodiments herein are not meant to be construed in a limiting sense. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. Furthermore, it shall be understood that all aspects of the invention are not limited to the specific depictions, configurations or relative proportions set forth herein which depend upon a variety of conditions and variables. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is therefore contemplated that the invention shall also cover any such alternatives, modifications, variations or equivalents. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

REFERENCE SIGNS LIST

100 liquid collector
110 flow passage member
111 outside tube
112 inside member
113 flow passage
113d flow passage
120 contact member
121 flow-passage-member receiving hole
123 contact surface
130 container
131 peripheral wall
132 bottom portion
133 opening
138 reservoir space
139 scale mark
200 liquid collector

210 flow passage member
 211 outside tube
 212 inside member
 213 flow passage
 213d flow passage
 220 contact member
 221 flow-passage-member receiving hole
 223 contact surface
 230 container
 231 peripheral wall
 232 bottom portion
 233 opening
 238 reservoir space
 239 scale mark
 300 liquid collector
 310 flow passage member
 311 outside tube
 312 inside member
 313 flow passage
 320 contact member
 321 flow-passage-member receiving hole
 323 contact surface
 330 container
 331 peripheral wall
 332 bottom portion
 333 opening
 338 reservoir space
 339 scale mark
 340 inner member
 341 cone-shaped portion
 400 liquid collector
 410 flow passage member
 411 outside tube
 412 inside member
 413 flow passage
 420 contact member
 421 flow-passage-member receiving hole
 423 contact surface
 425 sleeve
 430 container
 431 peripheral wall
 432 bottom portion
 433 opening
 438 reservoir space
 439 scale mark
 440 inner member
 441 cone-shaped portion
 500 liquid collector
 510 flow passage member
 511 outside tube
 512 inside member
 513 flow passage
 530 container
 531 peripheral wall
 532 bottom portion
 533 opening
 538 reservoir space
 539 scale mark
 540 inner member
 541 cone-shaped portion
 610 flow passage member
 613 flow passage
 615 groove
 710 flow passage member
 713 flow passage
 715 through hole
 715A through hole

715B through hole
 810 flow passage member
 812 single pipe
 812A single pipe
 812B single pipe
 813 flow passage
 815 hole
 816 gap
 880 flow passage member
 882 pipe
 883 flow passage
 885 hole
 910 flow passage member
 911 outer tube
 912 inner pipe
 912A inner pipe
 912B inner pipe
 913 flow passage
 915 hole
 916 gap
 917 outer peripheral gap
 999 liquid
 1000 surface
 1101 one end
 1102 other end
 1111 inner peripheral surface
 1121 groove
 1122 outer peripheral surface
 1211 inlet opening
 1212 outlet opening
 2101 one end
 2102 other end
 2111 inner peripheral surface
 2121 groove
 2122 outer peripheral surface
 2211 inlet opening
 2212 outlet opening
 2314 inner peripheral surface
 3101 one end
 3102 other end
 3111 inner peripheral surface
 3121 groove
 3122 outer peripheral surface
 3211 inlet opening
 3212 outlet opening
 3411 first opening
 3412 second opening
 4101 one end
 4102 other end
 4111 inner peripheral surface
 4121 groove
 4122 outer peripheral surface
 4211 inlet opening
 4212 outlet opening
 4411 first opening
 4412 second opening
 5101 one end
 5102 other end
 5111 inner peripheral surface
 5121 groove
 5122 outer peripheral surface
 5411 first opening
 5412 second opening
 6122 outer peripheral surface
 9111 inner peripheral surface

29

9122 outer peripheral surface

Ds pipe axis direction

Dv vertical direction

The invention claimed is:

1. A liquid collector for collecting liquid, the liquid collector comprising:

a flow channel member including a flow channel that connects one end and other end of the flow channel member;

a contact member disposed at the one end of the flow channel member, and that includes an elastic material; and the contact member comprising:

an inlet opening configured to come into contact with a surface of the target from which the liquid is to be collected and to allow introduction of the liquid that is on or near the surface, and

an outlet opening connected to the inlet opening to provide fluid communication and also connected to the flow channel member to provide fluid communication, wherein the one end of the flow channel member is fitted to the outlet opening of the contact member;

a container configured to receive the liquid discharged from the flow channel at the other end of the flow channel member; and

a cone-shaped portion disposed in the container, wherein at least a portion of the cone-shaped portion in the pipe axis direction has an inner diameter that decreases toward a bottom portion in the pipe axis direction, wherein the other end of the flow channel member is in contact with an inner peripheral surface of the cone-shaped portion.

2. The liquid collector according to claim 1, wherein the contact member includes a sleeve that has a tubular shape and extends toward the other end of the flow channel member.

3. The liquid collector according to claim 1, wherein the contact member has a contact surface in a region around the inlet opening, the contact surface coming into contact with the surface of the target from which the liquid is to be collected.

4. The liquid collector according to claim 3, wherein at least a portion of the contact surface is spherical.

5. The liquid collector according to claim 1, wherein the flow channel member includes a plurality of the flow channels that extend in a pipe axis direction connecting the one end and the other end of the flow channel member.

6. The liquid collector according to claim 5, wherein the plurality of the flow channels are substantially parallel to each other.

7. The liquid collector according to claim 5, wherein the flow channel member has a columnar shape that extends in the pipe axis direction connecting the one end and the other end of the flow channel member, and has a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the flow channel member.

8. The liquid collector according to claim 5, wherein the flow channel member includes

30

an outside tube having a tubular shape that extends in the pipe axis direction, and

an inside member inserted in the outside tube and having a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the flow channel member, and

wherein an inner peripheral surface of the outside tube and the plurality of grooves in the inside member define the plurality of the flow channels.

9. The liquid collector according to claim 5, wherein the flow channel member has a columnar shape that extends in the pipe axis direction, and the plurality of the flow channels extend through the flow channel member in the pipe axis direction to provide communication between the one end and the other end of the flow channel member.

10. The liquid collector according to claim 5, wherein the flow channel member includes a plurality of single pipes that are bundled together and that extend in the pipe axis direction.

11. The liquid collector according to claim 5, wherein the flow channel member includes:

an outer tube having a tubular shape that extends in the pipe axis direction, and

a plurality of inner pipes that are inserted in the outer tube and that extend in the pipe axis direction.

12. The liquid collector according to claim 5, wherein the container has an opening at a side opposite to a side at which the bottom portion is provided in the pipe axis direction, and

wherein the contact member is disposed to close the opening.

13. The liquid collector according to claim 1, wherein the container includes:

a peripheral wall having a tubular shape that extends in the pipe axis direction, and

a bottom portion that blocks the peripheral wall at one side in the pipe axis direction.

14. The liquid collector according claim 13, wherein the bottom portion of the container is spaced from the other end of the flow channel member in the pipe axis direction, and

wherein a reservoir space to store the liquid is defined between the bottom portion of the container and the other end of the flow channel member.

15. The liquid collector according claim 13, wherein the peripheral wall of the container has an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction.

16. The liquid collector according to claim 13, wherein the other end of the flow channel member is in contact with an inner peripheral surface of the peripheral wall of the container.

17. The liquid collector according to claim 13, wherein the container has an air hole that provides communication with an outside of the container.

18. The liquid collector according to claim 13, wherein the container has a scale mark to indicate an amount of the liquid received in the container.

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